S Vijay Malan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.90
SRI CHAITANYA TECHNO SCHOOL	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2024	93.80%
SRI CHAITANYA TECHNO SCHOOL	Class X (CENTRAL BOARD OF SECONDARY EDUCATION)	2022	92.40%

ACHIEVEMENTS

Solved **300+ Data Structures & Algorithms (DSA) problems** across coding platforms, strengthening problem-solving and algorithmic skills.

Achieved a **peak LeetCode Contest Rating of 1516**, demonstrating competitive programming proficiency and strong coding performance.

PROJECTS

Multimodal Agentic RAG | Github

Tech Stack: Python, RAG, LLMs, LlamaParse, PyMuPDF

- Developed a multimodal RAG pipeline for processing documents containing **text, images, and tables** and generating relevant responses from extracted information.
- Implemented **semantic and agentic chunking strategies** to organize document content into meaningful chunks for efficient retrieval and downstream processing.
- Built an agentic retrieval workflow incorporating **query rephrasing, semantic retrieval, and relevance assessment** to improve the selection of relevant document context.
- Integrated multimodal LLMs to process **images and tables and convert them into textual representations**, enabling their inclusion in the retrieval pipeline.
- Combined **document parsing, multimodal processing, retrieval, and LLM-based generation** into an end-to-end pipeline for question answering over complex documents.

Connecta - Video Conferencing Platform | Github

Tech Stack: React.js, Node.js, Express.js, MongoDB, WebRTC, Socket.io, Material UI

- Developed a real-time video conferencing platform enabling **multi-user video/audio calling, screen sharing, and in-meeting chat** with dynamic participant management.
- Implemented real-time communication using **WebRTC and Socket.io** for peer-to-peer media streaming, signaling, and participant coordination.
- Built WebRTC signaling using **SDP offer/answer negotiation, ICE candidate exchange, and STUN server integration** to establish peer connections.
- Designed RESTful APIs using **Node.js and Express.js** for user authentication and meeting management.
- Developed a responsive React interface using **Material UI** for meeting controls, participant management, and user interactions.

Stock Exchange Simulator | Github

Tech Stack: Next.js, React, TypeScript, Node.js, Express.js, MongoDB

- Developed a full-stack stock trading simulator supporting **virtual stock buying, selling, portfolio management,**

and transaction history through a responsive web application.
- Implemented RESTful APIs using **Node.js and Express.js** for user authentication, stock information, trading operations, and portfolio requests.
- Built a responsive trading dashboard using **Next.js and React** to display available stocks, holdings, account balance, and transaction information.
- Integrated **MongoDB** for persistent storage of user profiles, virtual balances, stock holdings, and transaction records.
- Built portfolio analytics to calculate **investment value, profit/loss, remaining account balance, and transaction history**.

SKILLS

Programming Languages: C, C++, Python, JavaScript, TypeScript

Frontend: HTML, CSS, React.js, Next.js, Material UI

Backend: Node.js, Express.js, REST APIs, Socket.io

Databases: MongoDB, MySQL

Developer Tools: Git, GitHub, Postman, Vercel

AI & Data Science: NumPy, Pandas, Scikit-learn, LlamaParse, PyMuPDF

RELEVANT COURSES

Computer Science: Data Structures & Applications, Introduction to Programming, Artificial Intelligence

Mathematics & Statistics: Elementary Linear Algebra, Probability & Random Variables, Biostatistics

Languages: Spoken Japanese Basics (Grade: A)

Programming & Data Science: R and Python for Biologists, Basic Bioinformatics, Big Data Biology & Biological Databases

EXTRACURRICULAR

Secured 2nd Place (Doubles) in **Table Tennis** at the Department Sports Day, IIT Hyderabad.

Secured 3rd Place in **Chess** at the Department Sports Day, IIT Hyderabad.

National Sports Organisation (NSO) - **Athletics**, IIT Hyderabad